Introduction to Economics

Class 12 — Perfect competition; monopoly begins

Pavel Kocourek

Universitat de Barcelona · Group A6

2 November 2026

Slide 1 Title

Block 4's second class. On Thursday the print shop's costs were taken apart: fixed and variable, the cost per bag, the cost of one more bag. Today those costs meet a price the shop cannot choose, and they decide how many bags it prints, when it prints none at all, and what happens over the years once anyone can open a shop next door. A real day of electricity prices opens the class and is explained after the break; the last two slides before the summary meet the opposite kind of firm, one that sets its own price.

Quiz 9

10 minutes · pens down at 10:10

Circle the letter of your answer in the block at the top.

A question marked + can have more than one correct letter: circle all of them.

Changed your mind? Cross out that circle and circle the new letter.

Write your name on the block, tear the strip off along the dashed line, hand in *only the strip*.

Quiz Start of class, on paper

A short quiz on paper opened the class, with questions on both sides of the sheet: circle the letter of your answer in the block at the top; a question marked + can have more than one correct letter; write your name on the block, tear the strip off along the dashed line and hand in only the strip.

Plan for today

- A firm that cannot choose its price: how much to produce
- When to produce nothing; the firm's supply curve
- A market's supply: a day of electricity prices
- The long run: entry and exit until economic profit is zero
- Monopoly: a first look

Slide 2 Plan for today

The class at a glance. Before the break, one firm: why the print shop takes its price as given, how it picks the number of bags from the cost of one more, when it is better to print nothing for a while, and how those answers add up to its supply curve. After the break, many firms: a market's supply built from its sellers, with a real day of electricity prices as the test; then what entry and exit do over the years; and, to end, the one seller with no rivals.

Two halves with a ten-minute break.

Warm-up: land, labour or capital?

Our print shop's inputs: land, labour or capital?

1. The workshop building it rents
   capital: a building is made by people
2. The plot of ground under the building
   land: nature's, not made by people
3. The printers' working hours
   labour: paid wages
4. Who owns and supplies all of these?
   households: as owners and as workers

Factors of production: land, labour, capital (class 3, slide 22).

Slide 3 Warm-up: land, labour or capital?

A warm-up on the factors of production, with the print shop as the example. The test for capital is whether people made the thing; for land, whether nature did; for labour, whether it is someone's time. The everyday word "rent" is the trap: the shop's monthly payment to its landlord mostly buys the use of a building, which is capital, while the economist's rent is only what the ground itself earns. The last item is Block 1's circular flow: every input is, in the end, owned or supplied by households.

Warm-up: who should make the summaries?

Exam prep: flashcards and summaries

In one hour — Ana: 6 flashcards or 2 summaries · Ben: 3 flashcards or 1 summary

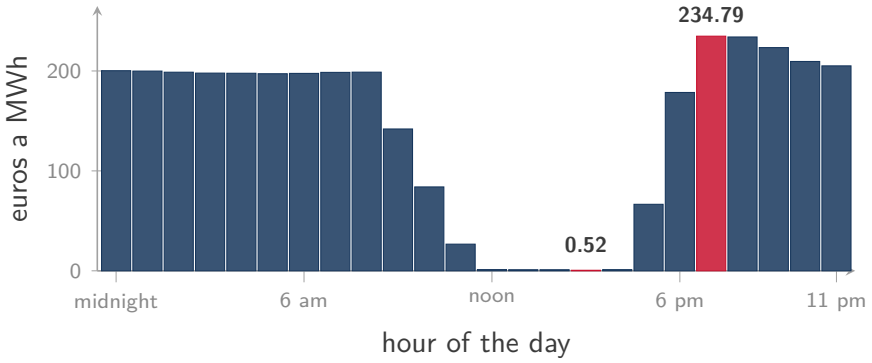
1. Can both gain by specialising?
   no: a summary costs each of them 3 flashcards
2. Ben gets faster: 3 flashcards or 2 summaries. Who makes summaries?
   Ben: a summary costs him 1.5 flashcards, Ana 3
3. After 2: at what price could they swap?
   between 1.5 and 3 flashcards a summary, e.g. 2

Comparative advantage: the lower opportunity cost (class 3, slide 18).

Slide 4 Warm-up: who should make the summaries?

A second warm-up, on comparative advantage. Being faster at everything is absolute advantage; a gain from specialising needs the two students to differ in what one summary costs them in flashcards, and in the first setting it costs both the same, so there is nothing to trade. Once one of them writes summaries at a lower cost in flashcards, that one should write them, and any swap price between the two opportunity costs leaves both better off. The lesson that carries into the rest of the class: what matters is what you give up, not how fast you are.

Sunday 11 October 2026: electricity at 0.52 euros a MWh at 3 pm, 234.79 at 7 pm



Same plants, same market, four hours apart: why?

OMIE (the Spanish–Portuguese wholesale electricity market), day-ahead price, mainland Spain, 11 Oct 2026; each bar averages the hour's four quarter-hour prices. 1 MWh = 1,000 kWh.

Slide 5 Sunday 11 October 2026: electricity at 0.52 euros a MWh at 3 pm, 234.79 at 7 pm

The hook: one wholesale electricity market, one Sunday, and a price that was about half a euro per megawatt-hour in mid-afternoon and some 450 times that four hours later, with the same power plants and the same buyers. Your bill at home swings far less: most contracts fix the price, and every bill adds grid charges and taxes on top of this wholesale price. Nobody fixes these numbers: they come out of supply meeting demand in each hour, and the day's tools explain both figures after the break. Keep the question open until then.

Perfect competition: our print shop is a price taker at 4 euros a bag

Tote printing in Barcelona:

- many shops, each one small
- the same printed bag at each
- anyone can open a shop

price taker: no shop can move the price



Each extra bag adds 4.00 to revenue: **marginal revenue** (MR) = price.

Competitive market: many sellers of the same good; none can move the price (class 5, slide 8).

Slide 6 Perfect competition: our print shop is a price taker at 4 euros a bag

Why the shop cannot set its price: with many small shops printing the same bag and nothing stopping a new one from opening, a customer quoted more than the going price walks next door, and a shop asking less only gives money away, since it can already sell all it prints at 4.00. The flat line is the demand this one shop faces, not the market's. The consequence carries the whole class: because the price does not move with the shop's output, each extra bag adds exactly 4.00 to revenue, so marginal revenue equals the price, and the shop's whole decision becomes one comparison, the price against the cost of the next bag. A seller that can choose its price comes at the end of the class.

At 4 euros a bag, profit peaks with four printers: 220 bags, 220 euros a day

printers	bags a day	total cost	MC a bag ΔTC ÷ Δbags	revenue 4.00 × bags	profit revenue − cost
0	0	180	—	0	−180
1	40	300	3.00	160	−140
2	120	420	1.50	480	60
3	180	540	2.00	720	180
4	**220**	**660**	**3.00**	**880**	**220**
5	240	780	6.00	960	180

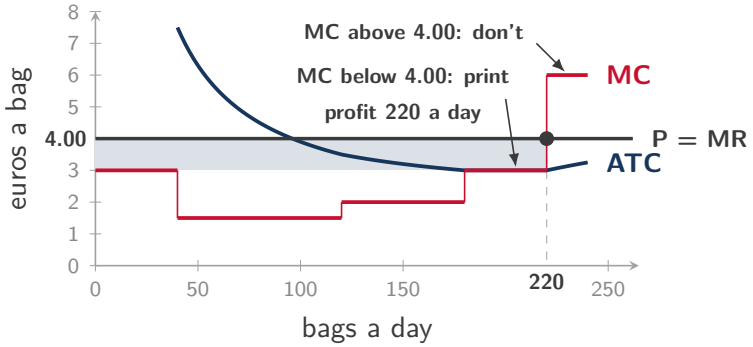
The fifth printer's 20 bags cost 6.00 each and sell for 4.00: profit falls.

Total cost (TC) and marginal cost (MC), the cost of one more bag: class 11, slide 10.
Costs here: those paid out.

Slide 7 At 4 euros a bag, profit peaks with four printers: 220 bags, 220 euros a day

Thursday's cost table with revenue and profit added. Profit here is revenue minus the costs actually paid out, so it starts at minus the rent and lease when nobody prints. The MC column, the cost of one more bag, is the reason the peak sits where it does: every bag up to 220 costs at most 3.00 to make and sells for 4.00, so each adds to profit, while the fifth printer's bags cost 6.00 each and lose 2.00 apiece. Not the first row where the cost per bag hits its low of 3.00, not the row with the most bags: the best row is where the next bag stops paying. The owner's own time stays out of this table because it is the same whatever she prints; it returns in the long run.

Print while marginal cost is below the price: 220 bags, profit 220 a day



Profit = (price − ATC) × bags = (4.00 − 3.00) × 220 = 220 a day.

MC: what one more bag adds to total cost · ATC, average total cost: total cost ÷ bags (class 11, slide 10).

Slide 8 Print while marginal cost is below the price: 220 bags, profit 220 a day

The same decision as a picture, and the rule the reading states: produce where marginal cost climbs past the price, which for a price taker means where P = MC, since MR = P. The reason is in the two regions: left of 220 each extra bag adds 4.00 of revenue and less than that of cost, so profit rises; right of it the next bag costs 6.00 and profit falls. What the rule is good for: the shop needs only the price and its own cost of one more bag, nothing about the market as a whole. The box is profit as the reading draws it: height price minus ATC, Thursday's total cost per bag, width the number of bags. Stopping at 180, where ATC first reaches its low of 3.00, would leave 40 bags unprinted that each earn 1.00.

In pairs, three minutes: a strawberry farm at the market price

A Maresme farm sells strawberries at the market price.
Extra cost of each further 100 boxes a day:

boxes a day	0→100	100→200	200→300	300→400	400→500
euros a box	4	5	7	10	14

1. Price 8 euros a box: how many boxes a day?
   300 boxes a day: the next 100 cost 10 each
2. The price rises to 12: how many now?
   400 boxes a day: a higher price, more boxes
3. At 8 euros, what do boxes 301–400 do to profit?
   profit falls by 200 euros: 100 × (8 − 10)

Slide 9 In pairs, three minutes: a strawberry farm at the market price

Exercise 1 is slide 8 on a farm, with the cost of each further hundred boxes in a table. At any price the farm picks every hundred that costs less than the price and stops at the first that costs more; a higher price makes more of the hundreds worth picking, which is why its answer to the price, its supply, slopes up. The last item shows what overshooting costs: a hundred boxes that fetch 8 each but cost 10 each do not add their revenue to profit, they subtract their loss. Picking everything, or stopping where the cost per box is lowest, are the two wrong rules.

Below 2 euros a bag the shop prints nothing: the rent is paid either way

price a bag	4.00	2.50	1.80
best if open	220 bags	180 bags	120 bags
revenue	880	450	216
printers' wages	480	360	240
rent and press	180	180	180
profit if open	+220	−90	−204
profit if shut down	−180	−180	−180
so	print	print	**shut down**

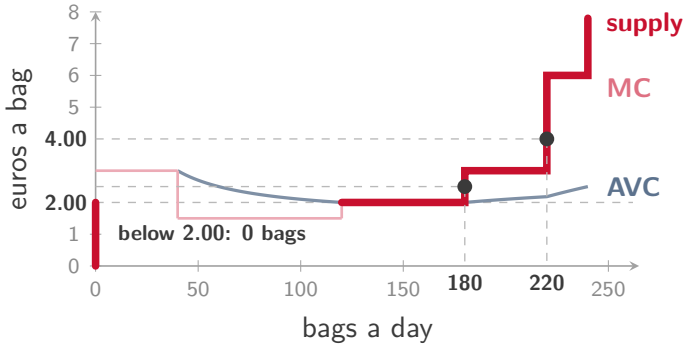
Shut down (for now) if the price is below AVC: here below 2.00.

AVC, average variable cost: printers' wages ÷ bags, lowest 2.00 · fixed: 180 whatever the output (class 11, slides 9–10) · sunk: already paid, gone whatever you choose (class 2, slide 23).

Slide 10 Below 2 euros a bag the shop prints nothing: the rent is paid either way

Each column first finds the best output by slide 8's rule and then compares two losses, open and shut down. The rent and lease, 180 a day, are owed either way this month: they are sunk, paid whatever she decides, so they cannot settle the question; what can is whether revenue covers the printers' wages, the variable cost. At 2.50 it does, with 90 to spare toward the rent, so staying open loses less than shutting down; at 1.80 it does not. Divided by bags, that is the rule: shut down when the price is below average variable cost, the wages per bag, whose lowest value is 2.00. It is why a hotel stays open at a loss in a quiet month.

The shop's supply curve: its rising marginal cost above the lowest AVC



A higher price, more bags: supply slopes up because MC rises.

AVC: printers' wages ÷ bags; MC: the cost of one more bag (class 11, slides 10 and 16).

Slide 11 The shop's supply curve: its rising marginal cost above the lowest AVC

Slides 8 and 10 drawn as one curve. At each price the shop prints where its MC staircase, the cost of one more bag, reaches the price, but only if the price is at least the lowest AVC, the wages per bag; below 2.00 it prints nothing, so the supply curve runs up the price axis and then along the staircase. The faint part of MC, where it is not above AVC, is never used. This is why class 5's supply curve slopes up, and what it is made of: the sellers' marginal costs, so anything that raises the cost of one more unit, wages, energy or a tax per unit, shifts supply up. The two dashed prices are slide 10's answers read off the curve.

A food truck's festival week: open today? (optional)

Pitch for the week: 600, paid, no refund · each day it opens: ingredients and a helper, 250

1. Sunday's sales would be 300. Open?
   yes: 300 covers the day's 250
2. Rain: Monday's sales would be 200. Open?
   no: opening loses 50 more
3. Does the 600 change either answer?
   no: the 600 is sunk, paid either way

Slide 12 A food truck's festival week: open today? (optional)

An optional exercise on slide 10's rule with an everyday firm. The pitch fee is paid and cannot be refunded, so it is sunk: it is the same whether the truck opens or stays shut, and it drops out of every day's decision. What is left is each day's revenue against each day's own cost, the ingredients and the helper; a day that covers them adds something toward the fee, a day that does not loses more by opening. Counting the fee against one day's sales is the mistake the exercise is built to catch.

Break

Back at 11:00

Problem set 2: due Thursday 5 November, on paper

Slide 13 Break

Ten minutes. Problem set 2 is due on Thursday 5 November, on paper, in class. After the break: the electricity question, then many firms instead of one.

Each plant runs while the price covers its extra cost of a MWh

	Solar farm	Gas plant
extra cost of a MWh	about 0	70
3 pm: price 0.52	**runs: 0.52 covers its cost**	**off: 0.52 is below 70**
7 pm: price 234.79 a little sun left	**runs: 234.79 covers its cost; little sun left to sell**	**runs: 234.79 covers 70**

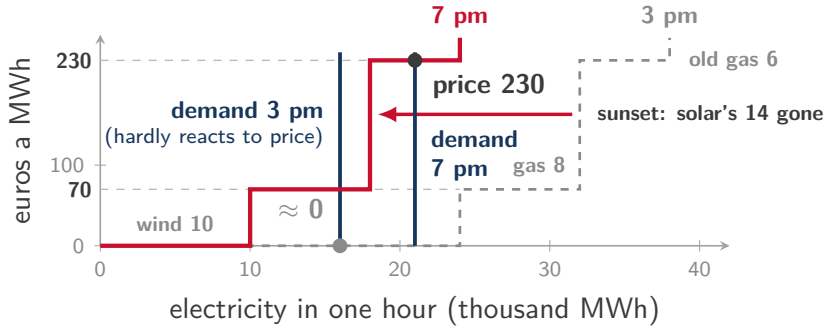
Panels, turbines, loans: paid either way, so they do not decide.

70 a MWh: gas plus the CO_2 permit (class 11, slide 14) · slide 10's rule: price against average variable cost; here every MWh costs the same, so that is the extra cost of a MWh.

Slide 14 Each plant runs while the price covers its extra cost of a MWh

The first half of the hook's answer. A power plant sells into a competitive market, one price for every MWh that hour, so slide 10's rule applies to it as it stands: run while the price covers the extra cost of a MWh, which for a plant whose every unit costs the same is also its average variable cost. The panels, the turbines and the loans are owed whether or not the plant runs, so they decide nothing; that is why the solar farm runs at half a euro and the gas plant does not, and why in the evening the gas plant runs while the solar farm has almost nothing to sell. The farm does not run because it is already paid for; it runs because one more MWh costs it about nothing.

Add up the plants: at 3 pm the cheapest are enough; at 7 pm the dearest is needed



Market supply: at each price, add the MWh every plant offers.

Own round numbers for one hour; real prices on slide 5 (OMIE). Each step is one kind of plant: its width the MWh it can offer, its height its extra cost of a MWh; 'old gas' is an older, dearer plant. Market supply adds the sellers' quantities at each price (class 5, slide 21).

Slide 15 Add up the plants: at 3 pm the cheapest are enough; at 7 pm the dearest is needed

The hook's payoff, and class 5's market supply made concrete. At each price, add the MWh every plant is willing to sell at that price; the prices are never added. The sum is horizontal because each plant's quantity is fixed by its own cost, so the market's quantity is simply their total; each step of the staircase is one kind of plant, its width what it can offer and its height its extra cost ('old gas': an older plant burning more gas per MWh), and demand within the hour barely reacts to the price, hence the vertical lines. In the afternoon the near-free plants alone cover what is needed, so the price falls to their cost; at sunset the solar block disappears, the staircase shifts left, and the evening's demand is met only if the dearest plants run, so the price rises to their cost. The dearest unit still needed sets the price: that is how the market operator sorts the offers each quarter-hour, and why more solar pushes midday prices down.

In pairs, three minutes: three egg farms, one market

Each farm's cost per dozen stays the same up to its limit.

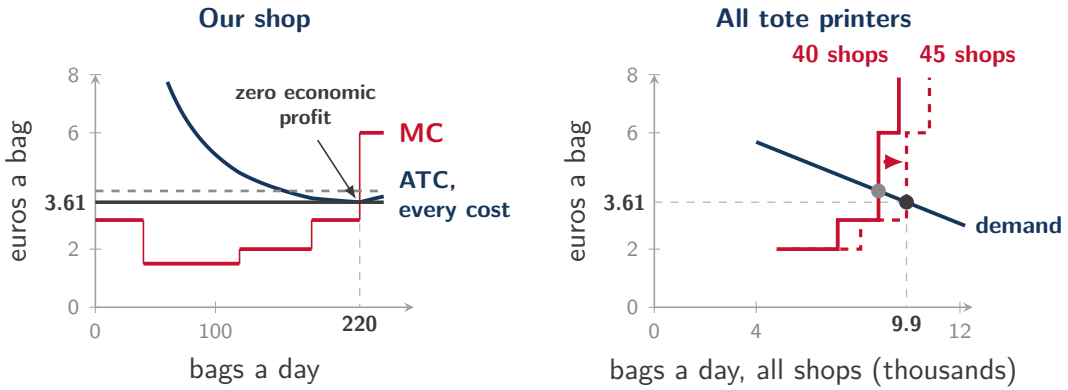
	cost per dozen	dozen a day (at most)
Farm A	1.00	300
Farm B	1.60	200
Farm C	2.20	100

1. Market supply at 2.00 a dozen?
   500 dozen: A and B; C stays shut
2. At 2.50?
   600 dozen: all three
3. At 1.20?
   300 dozen: only A
4. Buyers want 550 dozen a day at any price. The price?
   2.20: farm C is needed

Slide 16 In pairs, three minutes: three egg farms, one market

Exercise 2 is slide 15 with three farms. Market supply at a price is what every farm would sell at that same price, added together: at each price first decide who produces at all, since a farm whose cost per dozen is above the price stays out, then add their quantities; the prices and the costs are never added or averaged. The last item turns the question round: when buyers need more than the cheap farms can give, the price must rise to the cost of the dearest farm whose eggs are needed, which is the evening of slide 15 in a different market.

Profit draws new shops: the price falls until economic profit is zero



Entry stops when price = lowest ATC (every cost counted): 3.61.

Each shop: one press · every cost: class 11's costs plus the owner's 135 a day (class 11, slide 6)

long run: time for every cost to change (class 11, slide 18).

Slide 17 Profit draws new shops: the price falls until economic profit is zero

The long run begins, with every cost counted: Thursday's costs plus the owner's own 135 a day, the design job and the interest her savings gave up, which is what separates accounting from economic profit. Counting them lifts ATC; its lowest point, 3.61, is the price at which the shop pays her exactly what her best alternative would. At 4.00 she earns more, and because anyone can open a shop that gap is an invitation: new shops add their supply, the market curve moves right and the price falls until the gap is gone. Firms enter while the price is above ATC and exit while it is below, so the process can only stop where price equals ATC; since each shop also prints until its MC climbs past the price, that is the lowest ATC. The use of the result: in a competitive industry a profit is copied away, so a firm that wants to keep one must keep rivals out.

In the long run the price sits at the lowest ATC: market supply is flat at 3.61



More demand brings more shops, not a higher price.

Zero economic profit: the owner's accounts still show 2,700 a month — what her job and savings would pay.

Slide 18 In the long run the price sits at the lowest ATC: market supply is flat at 3.61

Where the price comes to rest. Any price above 3.61 is a profit someone will open a shop to take, and any price below it a loss that makes someone give up the lease and the press, so in the long run the price can only sit at the lowest ATC with every cost counted: the long-run market supply is flat there. More demand lifts the price for a while, as each shop climbs its own supply curve, but then brings new shops rather than a lasting rise; less demand makes some leave. Zero economic profit is not zero income: the owner covers every cost, her own time included, and her accountant still shows the 2,700 a month her job and savings would have paid, so she stays. Where firms' costs differ, as with the power plants, the long-run supply slopes up instead.

In pairs, three minutes: repair stalls in the long run

Phone-screen repair stalls, all alike. Lowest ATC, the owner's time counted: 40 euros a repair. Price today: 55.

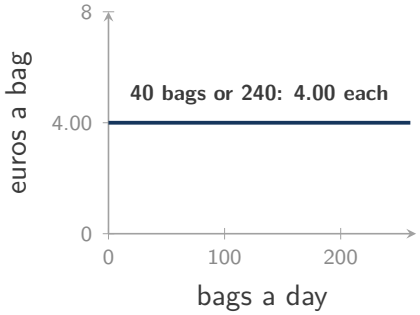
1. Economic profit today: positive, zero or negative?
   positive: 55 > 40, new stalls open
2. Where does the price go?
   down to 40, as new stalls add supply
3. At 40, do the owners quit?
   no: they earn their best alternative
4. A new 5-euro fee on every repair. Long-run price?
   45: stalls leave until the price covers 45

Slide 19 In pairs, three minutes: repair stalls in the long run

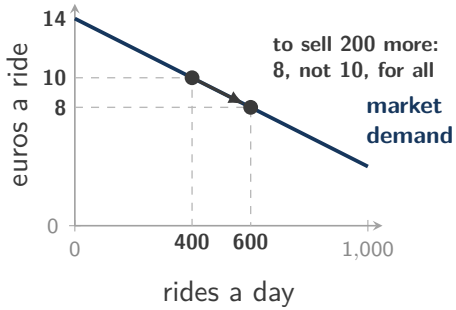
Exercise 3 is slides 17 and 18 on a new trade. With every cost counted, a price above the lowest ATC means economic profit, and economic profit draws entry until the price has fallen back to that ATC; at that point owners earn exactly their best alternative, which is a reason to stay, not to leave. A fee on every repair raises every stall's lowest ATC by the same amount, so at the old price stalls lose money, some leave, supply shrinks and the price rises until it covers the new ATC: in the long run the buyers pay the whole fee, even though in the short run the owners feel it.

A monopoly is the only seller: to sell more, it must cut its price

Our print shop: a price taker



A mountain's only cable car: a price maker



A price taker sells any amount at the price; a monopoly picks a point on the demand curve.

Which point it picks: Monday 9 November (reading Ch. 11).

Slide 20 A monopoly is the only seller: to sell more, it must cut its price

The print shop faces a flat demand because its customers have many other shops to go to; a seller with no rivals and no close substitute faces the whole market's demand, which slopes down. That is why it must cut its price to sell more, and for every rider, not only the extra ones: as the only seller, it is the market. So it chooses a point on the demand curve, a price and the quantity that goes with it, which makes it a price maker; what that is good for is that no rival pushes its price down to cost, as entry did on slide 17. It still cannot charge what it likes and sell the same amount: a higher price means fewer buyers. Which point it picks, and what that costs society, is Monday's question.

Rivals are kept out by a key resource, a law, or costs that fall with size

A key resource

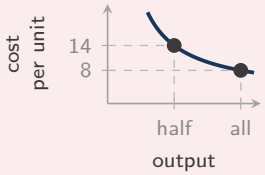
the only quarry of a rare stone

The law

a patent: no copies for 20 years

Falling costs: a natural monopoly

Spain's high-voltage grid: one company, Red Eléctrica



own numbers: one firm, 8 a unit; two firms, half each, 14

What keeps rivals out is a **barrier to entry**.

Economies of scale: ATC falls as output grows (class 11, slide 20).

European Patent Convention, art. 63 (20 years from filing); Ley 24/2013 del Sector Eléctrico, art. 34.2 (Red Eléctrica, transportista único).

Slide 21 Rivals are kept out by a key resource, a law, or costs that fall with size

Why a monopoly can last: something must stop rivals from entering, otherwise slide 17's entry would push the price down to cost here too. The three cards are three of the reading's four sources (the fourth: buying up rivals): one firm owns the input everyone needs; the law grants one firm the sole right, as a patent does for twenty years so that inventing pays; or one firm is simply the cheapest way to serve the whole market, because its cost per unit keeps falling as output grows, Thursday's economies of scale, so a second network would double the fixed cost for the same electricity. The grid that slide 15's competing plants share is such a natural monopoly, which is why grids, pipes and rail tracks are regulated rather than split. Size alone is not the test: a monopoly is a seller without a close rival.

Matcha gets popular: which curve moves? (optional)

Barcelona's matcha cafés: many, all alike; lowest ATC 3.50 a cup.

1. A video makes matcha fashionable. Which curve moves?
   demand shifts right
2. The price rises to 4.50. Does each café's supply curve shift?
   no: each café sells more, along its supply
3. New cafés open. What moves now?
   market supply shifts right; the price heads back to 3.50

Slide 22 Matcha gets popular: which curve moves? (optional)

An optional exercise on the difference between a shift and a move along a curve, using slides 17 and 18. A change in tastes changes how much people want at every price, so the demand curve itself moves. A higher price does not move any café's supply curve: each café simply sells more along the curve it already had. What does move market supply is new cafés, whose cups are added at every price, and that is what carries the price back toward the lowest ATC. Reading a higher price as a shift of supply is the phantom shift the exercise is built to catch.

One grid or two: positive or normative? (optional)

1. *"Two grids side by side would cost more per MWh carried."*
   positive: data can check it
2. *"The grid should belong to the state."*
   normative: a judgement of values
3. A and B both want cheap electricity. A: *"a state company would keep costs lowest."* B: *"a regulated private one would."* Values or a prediction?
   a prediction: how each would behave

Positive: a claim evidence can test · normative: a claim about what should be (class 4, slide 7).

Slide 23 One grid or two: positive or normative? (optional)

An optional exercise on positive and normative claims, hung on slide 21's grid. A claim that data could in principle check is positive, whether or not it turns out to be true; a claim with a "should" in it rests on values and is normative. The third item is the subtle case: two people who want the same thing and give opposite advice are not disagreeing about values but about how a state company and a regulated private one would behave, which is a prediction, a matter of evidence. Not every disagreement is about values, and a claim is not normative just because it is about policy.

A price taker produces while marginal cost is below the price.

Below the lowest average variable cost, it produces nothing: fixed costs are sunk.

Market supply adds the firms' quantities; in the long run economic profit is zero. A monopoly is the only seller: to sell more, it must cut its price.

Slide 24 In one line

The class in four lines, to read slowly: the first is slides 6 to 9, the second slides 10 and 11, the third slides 14 to 19, and the fourth slides 20 and 21.

Before Thursday 5 and Monday 9 November

- Thursday 5 November: poster session, the full two hours; one presenter per group, the whole group answers questions
- Bring your poster printed, A2 portrait; e-mail the PDF by Wednesday 4 November, 8 pm
- Problem set 3 (Block 4): handed out today; due Thursday 12 November, on paper, in class
- Monday 9 November: monopoly; oligopoly and the prisoner's dilemma — read Mankiw & Taylor Ch. 11, pp. 242–261 and Ch. 13, pp. 280–296
- Practise in Sextant: the units the course page lists

pkocourek@ub.edu · pkocourek.com/teaching/intro-economics · pkocourek.com/sextant/?c=intro-econ

Slide 25 Before Thursday 5 and Monday 9 November

Thursday 5 November is the poster session, the full two hours: one presenter per group, the whole group answers questions; bring the poster printed, A2 portrait, and e-mail the PDF by Wednesday 4 November, 8 pm. Problem set 3, on Block 4, is handed out today and due on Thursday 12 November, on paper, in class; Problem set 2 is due on Thursday 5 November, on paper, at the poster session. Monday 9 November: monopoly, then oligopoly and the prisoner's dilemma. Reading: Mankiw & Taylor chapter 11, pages 242 to 261, and chapter 13, pages 280 to 296. Practise in Sextant with the units the course page lists.

pkocourek.com/teaching/intro-economics · pkocourek@ub.edu · pkocourek.com/sextant/?c=intro-econ